SHUBHA DIXIT

<u>Address for Correspondence</u>:
149 D, Vill + Post – Bhikhepur,
Auraiya, U.P. - 206121
Email: shubha.dixit.9@gmail.com

CAREER OBJECTIVE

To establish myself as an independent researcher in the field of environmental microbiology and biotechnology by advancing sustainable solutions for environmental pollution and agricultural challenges through innovative microbial technologies. My research focuses on plant–microbe interactions, fungal endophytes, rhizospheric microbiology, microbial ecology, and arsenic biotransformation and bioremediation to develop environmentally sustainable strategies for contaminant remediation and enhanced plant resilience. I aspire to integrate classical microbiology with advanced molecular, genomic, and multi-omics approaches to unravel microbial functions and their applications in environmental and agricultural systems. Through high-quality, interdisciplinary research, scientific collaborations, and continuous learning, I seek to contribute to impactful publications, translational research, and the advancement of scientific knowledge while building a career in academia and research dedicated to sustainable environmental development.

EMPLOYMENT EXPERIENCE

<u>National Institute of Hydrology, Roorkee, Uttarakhand</u> **(January 2022 – January 2024)**

Worked as a Resource Person in the research project **"Understanding Arsenic Mobilization in the Groundwater of Haridwar and Formulating Remediation Measures,"** contributing to groundwater quality assessment and arsenic contamination studies. Responsibilities included conducting field sampling and groundwater collection, performing physicochemical and environmental analyses of water samples, designing and executing laboratory experiments, operating analytical instruments following QA/QC protocols, and interpreting experimental data using statistical approaches. Contributed to technical reporting, project documentation, and multidisciplinary research efforts aimed at understanding arsenic mobilization mechanisms and developing sustainable remediation strategies for arsenic-contaminated groundwater.

AREA OF INTREST

❖ **Microbiology** – Environmental Microbiology, Microbial Ecology, Rhizosphere and Endophytic Microbiology, Microbial Genetics, Mycology, Bacteriology, Microbial Physiology

❖ **Biotechnology** – Environmental Biotechnology, Bioremediation, Plant–Microbe Interactions, Molecular and Cellular Biology, Environmental Toxicology, Sustainable Agriculture

❖ **Environmental Sciences** – Water Quality Assessment, Groundwater Microbiology, Arsenic Biogeochemistry, Heavy Metal Biotransformation, Environmental Monitoring, Pollution Control

❖ **Research Techniques** – Molecular Microbiology, Microbial Genomics, Multi-omics, Microbial Diversity Analysis, Bioinformatics, Analytical Instrumentation

ACADEMIC PROFILE

Year(s)	Qualification – Degree / Diploma/ Certificate	Board/University	College / Institute	Percentage / CGPA
2024	PhD (Biological Science)	AcSIR	CSIR-Indian Institute of Toxicology Research, Lucknow	Ongoing
2020-22	M.Sc. (Applied Microbiology	BANASTHALI	BANASTHALI	9.00 CGPA

	and Biotechnology)	VIDYAPITH	VIDYAPITH	
2017-20	B.Sc. (Bioscience)	BANASTHALI VIDYAPITH	BANASTHALI VIDYAPITH	7.45 CGPA
2017	INTERMEDIATE	CBSE	ST. FRANCIS ACADEMY	70%
2015	HIGH SCHOOL	CBSE	ST. FRANCIS ACADEMY	7.6 CGPA

ACHIEVEMENTS AND CERTIFICATIONS

- Secured First rank and gold Medal in M.Sc. AMBT batch of 2020 – 2022
- A recipient of a University Merit Certificate (Awarded to the students of BANASTHALI VIDYAPITH for making a scholastic grade point average of 9 CGPA).
- Consistent performer in semester exams (CGPA OF 9.0 + during all the semesters of M.Sc.)
- University Representative in 10th Bhartiya Chaatra Sansad, New Delhi. (2020)
- Qualified NTA PhD entrance exams 2024 in Biotechnology and Microbiology

FELLOWSHIP AND AWARDS

- **DST-INSPIRE** FELLOWSHIP in Lifescience
- **AIR-199 ICAR AICE- JRF/SRF 2023**

MAJOR PROJECTS AND INTERNSHIP

M.Sc. Dissertation (Jan 2022 – July 2022) **(National Institute of Hydrology, Roorkee, Uttarakhand)**

Project Title: "The Evaluation of Constructed Wetland in Chemical and Bacterial Removal"

Description: Conducted a research project to evaluate the efficiency of constructed wetlands in improving water quality through the removal of chemical contaminants and microbial pollutants. The study involved the analysis of key physicochemical and environmental parameters influencing treatment performance, along with the isolation, cultivation, and identification of bacterial communities associated with the wetland system to investigate their role in pollutant degradation and ecological functioning. Standard microbiological and analytical techniques were employed for water quality assessment and bacterial characterization.

SKILLS

Research & Laboratory Skills: Environmental microbiology, microbial isolation and characterization, fungal and bacterial culturing, aseptic techniques, plant tissue culture, DNA extraction, PCR, agarose gel electrophoresis, ELISA, cell-based assays, microscopy, water quality analysis, spectrophotometry, Total Organic Carbon (TOC) analysis, Total Kjeldahl Nitrogen (TKN) estimation, ion chromatography, biochemical characterization of microorganisms, media preparation, and experimental design.

Analytical & Technical Skills: Research methodology, scientific writing, data analysis and interpretation, statistical analysis, environmental monitoring, laboratory quality assurance/quality control (QA/QC), method validation, and documentation.

Software & Bioinformatics: Microsoft Office Suite (Word, Excel, PowerPoint), Origin, BLAST, ArcGIS, QGIS, Google Earth Pro, and literature management tools.

Professional Competencies: Scientific problem-solving, critical thinking, hypothesis development, experimental planning, teamwork and collaboration, leadership, project management, time management, adaptability, attention to detail, responsibility, and effective scientific communication

SHORT-TERM TRAINING PROGRAM/CERTIFICATES WORKSHOP

- 5 days of Training on "Water and Wastewater Treatment" organized by the National Institute of Hydrology, Roorkee.
- 5 days of hands-on Training on "Water Quality Monitoring & Management" organized by the National Institute of Hydrology, Roorkee.
- 5 days of hands-on Training on "Environment Data Processing" organized by the National Institute of Hydrology, Roorkee.
- Participated in "2nd Roorkee Water Conclave 2022", organized by the Indian Institute of Technology, Roorkee
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CONFERENCES AND SYMPOSIUMS

International Conferences: 3; National Symposium: 1

International Conferences:

- Dixit, S International Conference on Future of Water Resources (ICFWR-2024) at Indian Institute of Technology Roorkee on 18-24 January 2024.
- 3rd Roorkee Water Conclave at Indian Institute of Technology Roorkee on 3-6th March 2024.

  - Dixit, S., Maurya, A., Singh, R., & Kumar, M. Assessment of Heavy Metal Contamination in Agriculture Soils of Laksar, Uttarakhand, (2025). Emerging Approaches in Risk Analysis and Translational Aspects of Health and Environment (EARTH 2025) held on November 12-15, 2025, at CSIR- Indian Institute of Toxicology Research, Lucknow

  - Dixit, S., Maurya, A., Singh, R., & Kumar, M. Spatial Distribution and Mapping of Heavy Metals in Agricultural Fields of Laksar, Uttarakhand (2026). Roorkee Water Conclave (RWC 2026) held on February 23-25, 2026, at the Indian Institute of Technology Roorkee and the National Institute of Hydrology, Roorkee.

  - Dixit, S., Maurya, A., Singh, R., Singh, S., and Kumar, M.: Microbe-Assisted Remediation Potential in Arsenic-Impacted Agricultural Soils of Laksar, Uttarakhand, EGU General Assembly 2026, Vienna, Austria, 3–8 May 2026, EGU26-811, https://doi.org/10.5194/egusphere-egu26-811, 2026

  - Dixit, S., Maurya, A., Nayak, S., Singh, S., Singh, R., & Kumar, M. From Arsenic Hotspots to Biological Solutions: Microbial Remediation of Contaminated Agricultural Soils of Laksar (2026). International Symposium on Emerging Contaminants in Water with Special Focus on Arsenic and Fluoride Contamination (ECWAF 2026) held on May 29-30, 2026, at the Indian Institute of Technology Roorkee.

  - Nayank, S., Dixit, S., & Kumar, M. Investigating the role of *Serendipita indica* and Endofungal Bacterium 06_IITR Association in cadmium Restriction in Tobacco (2025). Emerging Approaches in Risk Analysis and Translational Aspects of Health and Environment (EARTH 2025) held on November 12-15, 2025, at CSIR-Indian Institute of Toxicology Research, Lucknow.

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National Symposium:

- 10th Bharatiya Chhatra Sansad organized by MIT World Peace University on 20-23rd February 2020

PUBLICATIONS AND ABSTRACTS

- Dixit, S., Singh, R., (2024). Evaluation of Constructed Wetland for Chemical and Bacterial Removal**, ICFWR 2023**, **Jan. 18-20, 2024**, Roorkee, India.

- Oral presentation on "Evaluation of Constructed Wetland for Chemical and Bacterial Removal" in 3rd Roorkee Water Conclave at Indian Institute of Technology Roorkee 3rd- 6th March, 2024

- **Dixit, S.,** Maurya, A., Singh, A., Verma, S., Singh, R., & Kumar, M. (2025). Microbes' and Microplastics' Interactions in Freshwater Ecosystems: Fate and Implications for Environmental Health. In *Occurrence,* *Detection, and Fate of Microplastics in Freshwater Ecosystems* (pp. 1-39). Singapore: Springer Nature Singapore. 10.1007/978-981-95-0225-7_1

- Shukla, J., Singh, S., **Dixit, S.,** Kushwaha, A. S., Mohd, S., Saji, J., & Kumar, M. (2025). Efficient Removal of Arsenic from the Environment by Endophytic Fungus *Serendipita indica* through bio-accumulation in cells and adsorption on the cell wall. *Environmental Technology & Innovation*, 104229. https://doi.org/10.1016/j.eti.2025.104229

- Maurya, A., **Dixit, S.**, Nayak, S., Singh, S., & Kumar, M. (2026). Applications of Fungi in Biodegradation of Polymeric Waste and Micropollutants. In *Fungi in Wastewater Treatment: Volume 2* (pp. 101-123). Cham: Springer Nature Switzerland. https://doi.org/10.1007/978-3-032-01955-4_5

- Singh, S., Nayak, S., Maurya, A., **Dixit, S.,** & Kumar, M. (2026). Applications of Fungi in Wastewater and Innovative Approaches to Residual Waste Processing. In *Fungi in Wastewater Treatment: Volume 2* (pp. 283-308). Cham: Springer Nature Switzerland. https://doi.org/10.1007/978-3-032-01955-4_13

- Nayak, S., Singh, S., Maurya, A., **Dixit, S.,** & Kumar, M. (2026). Applications of Fungi in Remediation of Radioactive Contaminants. In *Fungi in Wastewater Treatment: Volume 2* (pp. 217-242). Cham: Springer Nature Switzerland. https://doi.org/10.1007/978-3-032-01955-4_10

HOBBIES AND EXTRA CO-CURRICULAR ACTIVITIES

- Badminton.
- Knowledge of basic Vedic Scriptures
- Photography, Poetry writing.
- Volunteering in National Service Scheme.

PERSONAL DETAILS

Date of Birth	:	9 March 1999
Father's Name	:	Mr. Brijendra Prasad Dixit
Mother's Name	:	Mrs. Seema Dixit
Gender	:	Female
Permanent Address	:	149 D, Vill + Post – Bhikhepur, Auraiya, U.P. - 206121
Mobility	:	Willing to relocate anywhere in India and overseas.

I declare that the details above are correct and true to the best of my knowledge.

SHUBHA DIXIT